VIKAS SHARMA

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EDUCATION

Cluster University of Jammu — Jammu, J&K, India
Bachelor of Computer Applications (BCA) — *Aug 2024 – May 2027*

TECHNICAL SKILLS

Languages: C++, JavaScript, TypeScript, Python, Java, SQL
Frontend: Next.js, React.js, Vite, Tailwind CSS, HTML5/CSS3
Backend & Systems: Node.js, Express.js, REST APIs, Socket.io, JWT, Concurrency, Linux
Databases: MongoDB, PostgreSQL, MySQL
Networking & Protocols: TCP/IP, UDP, OSI, TLS Handshakes, DNS, ARP, Subnetting
Cloud & Tools: Docker, AWS, Nginx, Git, GitHub, Postman
CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, DBMS, OS

EXPERIENCE

Full Stack Developer (Summer Internship) — Jun 2026 – Jul 2026
CodTech IT Solutions — *Hyderabad, Telangana (Remote)*

* Engineered backend microservices deployed via zero-downtime CI/CD pipelines, supporting persistent WebSocket connections via Socket.io for low-latency synchronization.
* Designed normalized MongoDB Atlas schemas and query pipelines with Mongoose, supporting efficient data access patterns across active sessions and user state.
* Refactored core service modules and resolved authentication and state-sync bugs, hardening endpoints with JWT-based role verification.

PROJECTS

PacketLens | *C++, Multi-threading, TCP/IP, TLS/HTTPS, PCAP Parsing* — 2026

* Engineered a multi-threaded Layer 2–7 Deep Packet Inspection engine in modern C++ to capture, parse, and filter high-volume network traffic from binary PCAP streams.
* Architected a concurrent worker pipeline using 5-tuple consistent hashing, routing session flows to dedicated Fast Path threads to eliminate flow-table race conditions without global lock contention.
* Implemented deep packet inspection across TLS Client Hello frames to extract plaintext SNI and Host headers, classifying encrypted web applications prior to symmetric session key exchange.
* Integrated real-time flow tracking and domain/IP blacklisting rules to drop unauthorized sessions and stream sanitized packets to an output PCAP file.

CodeSync | *React, Vite, Node.js, Express, Socket.io, Monaco Editor* — 2026

* Architected a real-time collaborative code editor supporting multi-user concurrent sessions synchronized through WebSocket-based event pipelines.
* Integrated OnlineCompiler API to support sandbox code compilation and execution across Python, Java, C++, and TypeScript.
* Optimized active-user state tracking per collaborative room using server-side key-value memory mapping to prevent stale connection leaks.

Retroleaf | *React, Node.js, Express, MongoDB, JWT, Tailwind CSS* — 2026

* Built a secure full-stack journaling platform with responsive UI, retro CRT rendering, and granular per-user private journal isolation.
* Engineered MongoDB aggregation queries to power interactive mood tracking and streak consistency analytics over custom time windows.
* Secured all state-modifying endpoints using stateless JWT authentication middleware and sanitized data inputs to prevent cross-account injection.

ACHIEVEMENTS & CODING PROFILES

LeetCode: Solved **200+** Data Structures & Algorithms problems covering Arrays, Graphs, Linked Lists, Trees, and Dynamic Programming.

Technical Profiles: Active contributor on GitHub with fully deployed open-source systems and full-stack projects.